

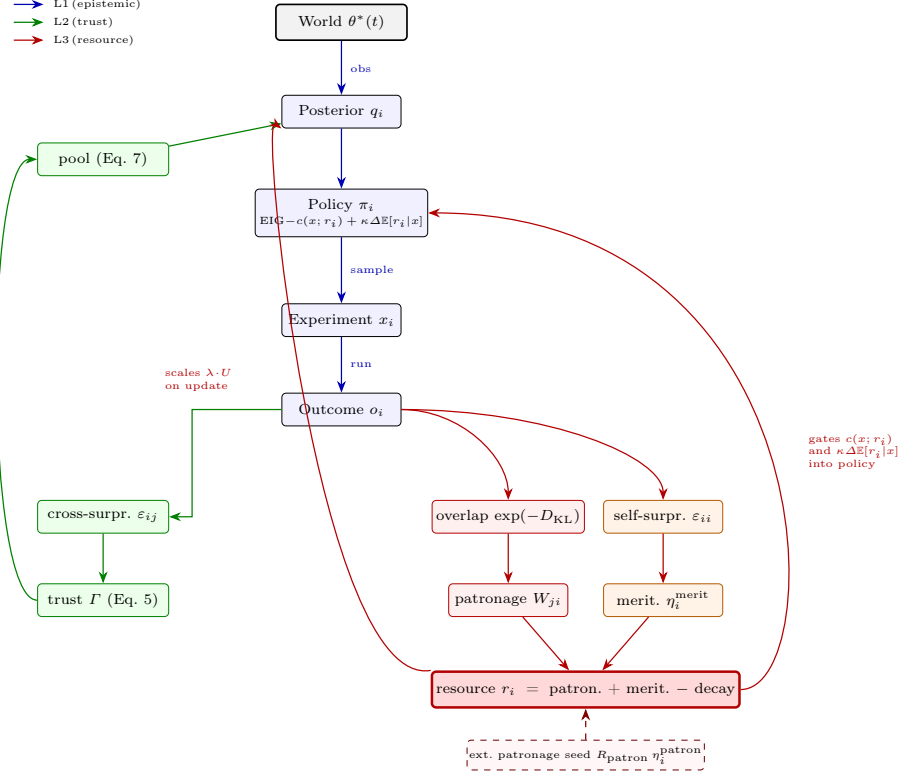


Fig. 1. The three closed feedback loops of the coupled belief–resource system, sharing one world–agent–policy–action spine. **L1 (blue, epistemic)**: standard single-agent active inference, world $\theta^* \rightarrow$ posterior $q_i \rightarrow$ policy $\pi_i \rightarrow$ experiment $x_i \rightarrow$ outcome $o_i \rightarrow$ posterior. **L2 (green, trust)**: outcomes generate cross-surprisals ε_{ij} which update the trust matrix Γ (Eq. 5); trust-weighted pooling (Eq. 7) closes the loop back into q_i . **L3 (red, resource)**: outcomes feed two parallel scoring channels — belief overlap drives the patronage flow W_{ji} (“allocate to those whose beliefs you share”), and self-surprisal ε_{ii} drives the meritocratic inflow share η_i^{merit} (“predictions borne out earn resources”). Both feed the resource state r_i , alongside an external patronage seed (dashed) controlled by the funding-agency policy. The resource state then gates two entry points back into the agent’s dynamics: a single arrow into the policy carries both the resource-gated experiment cost $c(x; r_i)$ and the expected-resource-gain term $\kappa \Delta \mathbb{E}[r_i | x]$, and a second arrow into the posterior update scales the λ -weighted socially-induced utility U pulling q_i toward the beliefs of resource-rich neighbours. **No loop closes on its own;** each closes through the others.